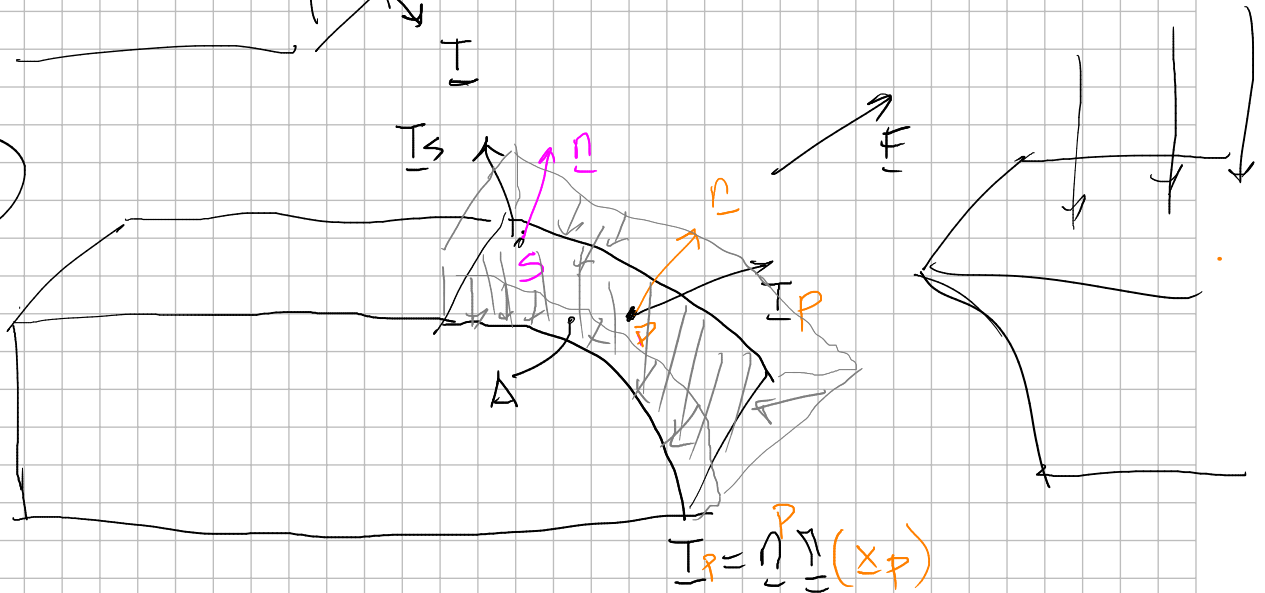
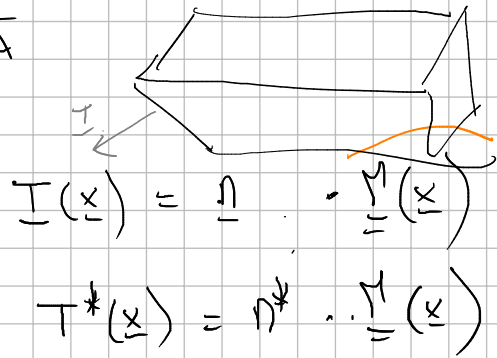
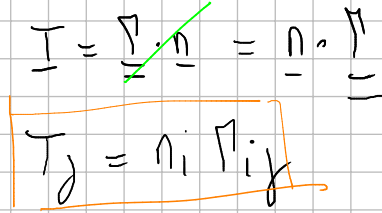


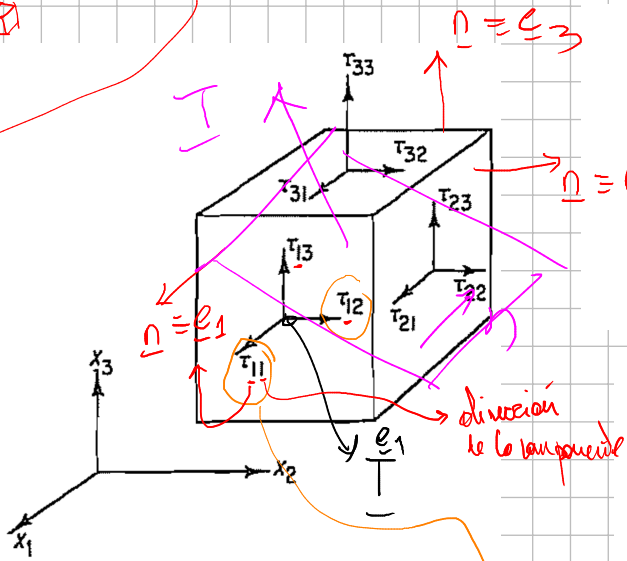
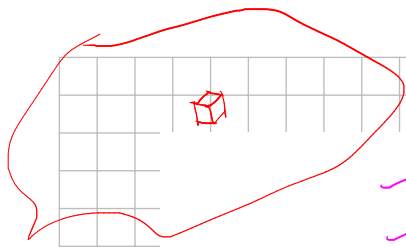


$$F = \int_A T(x) dA$$

$$I_S = \bigcap_{i=1}^n I_i(x_i)$$

↳ more sub-particles

$$\|n^s\| = 1$$



τ_{ij}

$$\begin{pmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{pmatrix}$$

$$\begin{pmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{pmatrix}$$

$$\underline{\underline{e_1}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{pmatrix} \tau_{11} & \tau_{12} & \tau_{13} \\ \tau_{21} & \tau_{22} & \tau_{23} \\ \tau_{31} & \tau_{32} & \tau_{33} \end{pmatrix} = \begin{bmatrix} \tau_{11} \\ \tau_{12} \\ \tau_{13} \end{bmatrix}$$

$$[\underline{\underline{e_1}}]^T \begin{pmatrix} \tau_{11} & \tau_{12} & \tau_{13} \\ \tau_{21} & \tau_{22} & \tau_{23} \\ \tau_{31} & \tau_{32} & \tau_{33} \end{pmatrix} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{pmatrix} \tau_{11} & \tau_{12} & \tau_{13} \end{pmatrix}$$

$$\left(\underline{\underline{e_1}} \right)_i = \tau_{ij} \rightarrow$$

$$\underline{\underline{\tau}} = \underline{\underline{P}} \underline{\underline{\tau}} \underline{\underline{P}}^T$$

$$\tau_{ij} = P_{im} P_{jn} \tau_{mn}$$

la transformación de coordenadas
 $\downarrow$
 tensor cartesiano de 2do orden